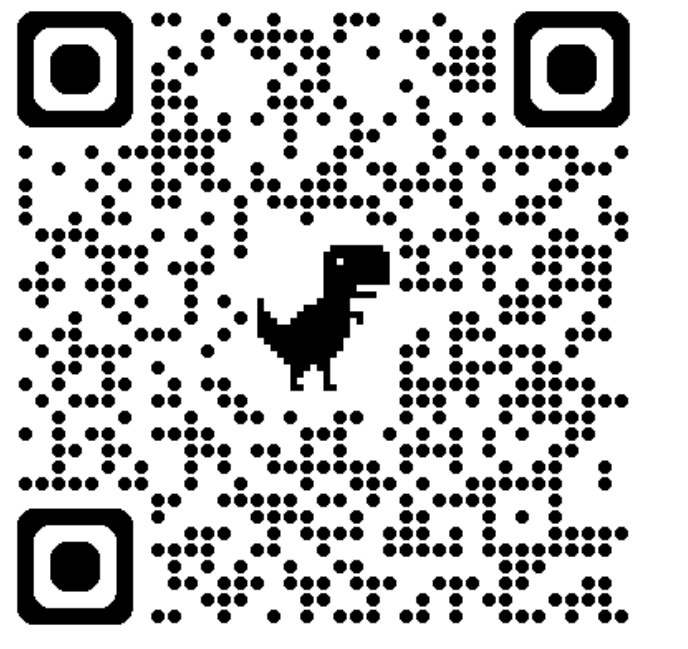


Symmetry and anomaly in lattice gauge theory

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[arXiv:2210.12967, 2302.13466, 2303.10977, 2304.11813, 2304.14815, 2401.01331, 2403.03420]



GitHub:

Introduction

- **Symmetry**: universal and fundamental tool in physics
- Structure in Nature: Lorentz symmetry, **Gauge invariance**, etc.
 - Fundamental forces, e.g., photon: $U(1)$ sym

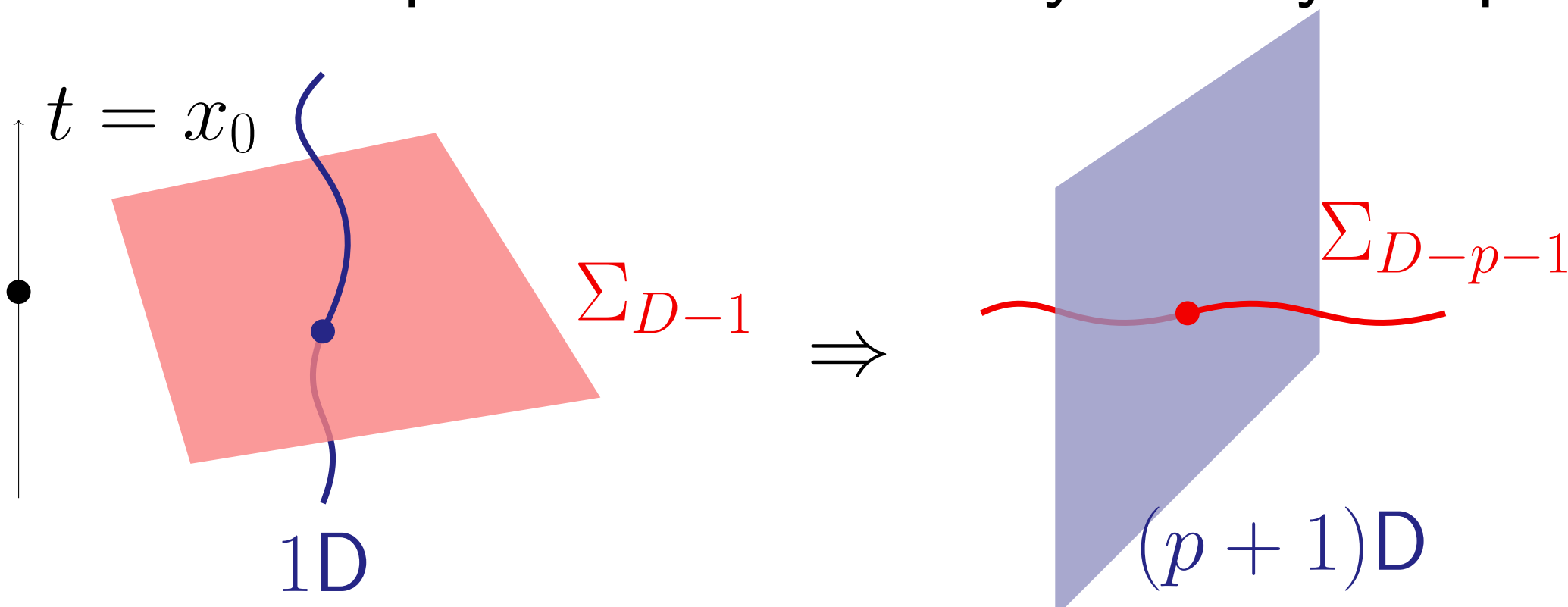


- Noether theorem and conservation law

$$\text{Sym} : \phi \mapsto \phi'; S(\phi) = S(\phi'), \quad \partial_\mu j^\mu = 0$$

$$\text{Charge } Q \equiv \int j^0 d^{D-1}x \text{ (time independent)}$$

- Recent viewpoint of notion of symmetry: Topological for $\Sigma \rightarrow \Sigma'$



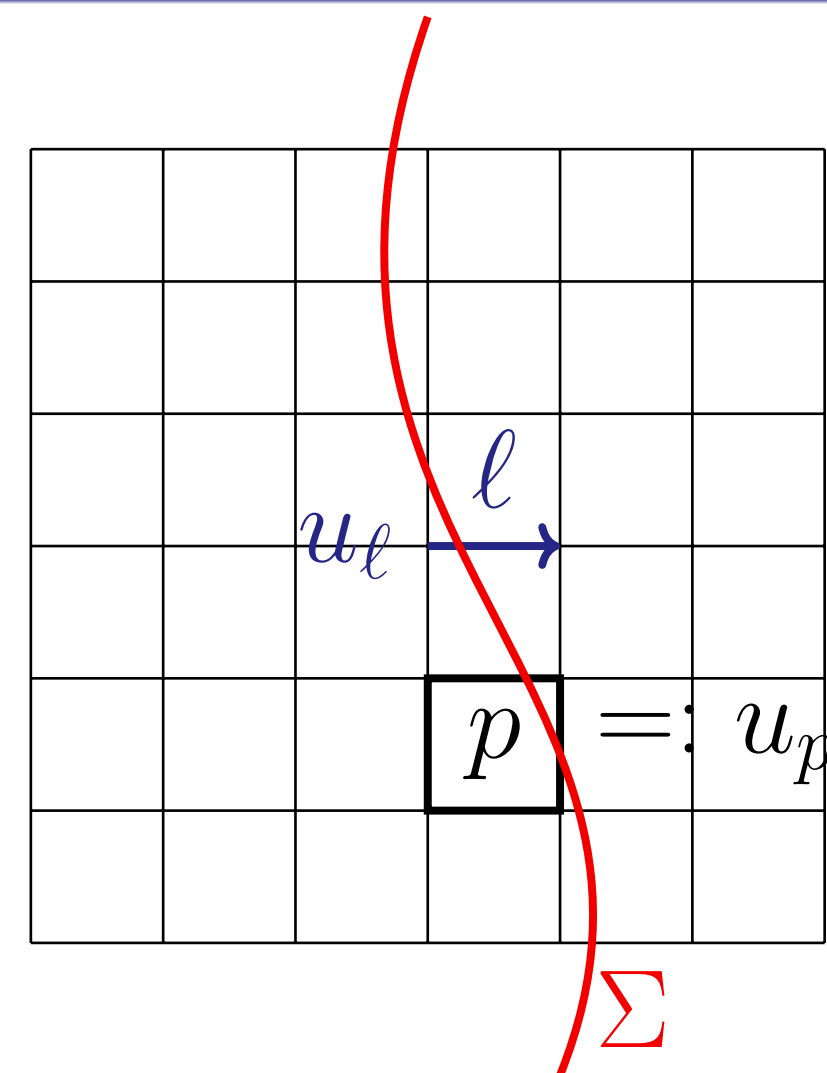
“Generalized symmetry”: 0-form $\Rightarrow$ p -form symmetry

[Gaiotto–Kapustin–Seiberg–Willet '14]

- This provides more restrictions on **low-energy dynamics** in QFT
- We will consider it on the basis of lattice gauge theory

Lattice gauge theory & numerical study

- Lattice regularization of QFT is a non-perturbative framework
- Lattice $SU(N)$ YM theory
 - Link variable $u_\ell \in SU(N)$ on link ℓ
 - Plaquette action $S \sim \sum_p \text{tr } u_p$
- Gauge symmetry: $u_\ell \mapsto \omega_n u_{\ell=(n,\mu)} \omega_{n+\hat{\mu}}^\dagger$
- **1-form center symmetry**: $\mathbb{Z}_N^{[1]}$



$$e^{\frac{2\pi i k}{N}} \in \mathbb{Z}_N \subset SU(N); \quad u_\ell \mapsto e^{\frac{2\pi i k}{N} \#(\Sigma, \ell)} u_\ell$$

Intersection number $\#(\Sigma, \ell) \in \mathbb{Z}$; Now $u_p \mapsto u_p$

Gauging generalized global symmetry

- Gauging the center symmetry

$$S \sim \sum_p \text{tr } e^{-\frac{2\pi i}{N} B_p} u_p \quad B_p: \text{2-form gauge field assoc. } \mathbb{Z}_N^{[1]}$$

invariant under $u_\ell \mapsto e^{\frac{2\pi i}{N} \lambda_\ell} u_\ell, B_p \mapsto B_p + (d\lambda)_p$

- cf. 't Hooft twisted boundary condition [79]:

$$u_{n+L\hat{\nu}, \mu} = g_{n, \nu}^{-1} u_{n, \mu} g_{n+\hat{\mu}, \nu}^{-1} \quad g_{n+L\hat{\nu}, \mu} g_{n, \nu}^{-1} g_{n, \mu} g_{n+L\hat{\mu}, \nu} = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N$$

't Hooft flux $z_{\mu\nu} = \sum B_p \bmod N$

Fractionality of topology is expected: $\propto 1/N$ [van Baal '82]

- We prove fractionality in terms of B_p !

- Lattice field theory: Most well-established computational approach
 - Numerical simulation under 't Hooft twisted b.c.

[Edwards–Heller–Narayanan, de Forcrand–Jahn,

Fodor–Holland–Kuti–Nógrádi–Schroeder, Kitano–Suyama–Yamada, Itou, ...]

- Based on LatticeQCD.jl by Y. Nagai & A. Tomiya (periodic b.c.)

[https://github.com/akio-tomiya/LatticeQCD.jl],

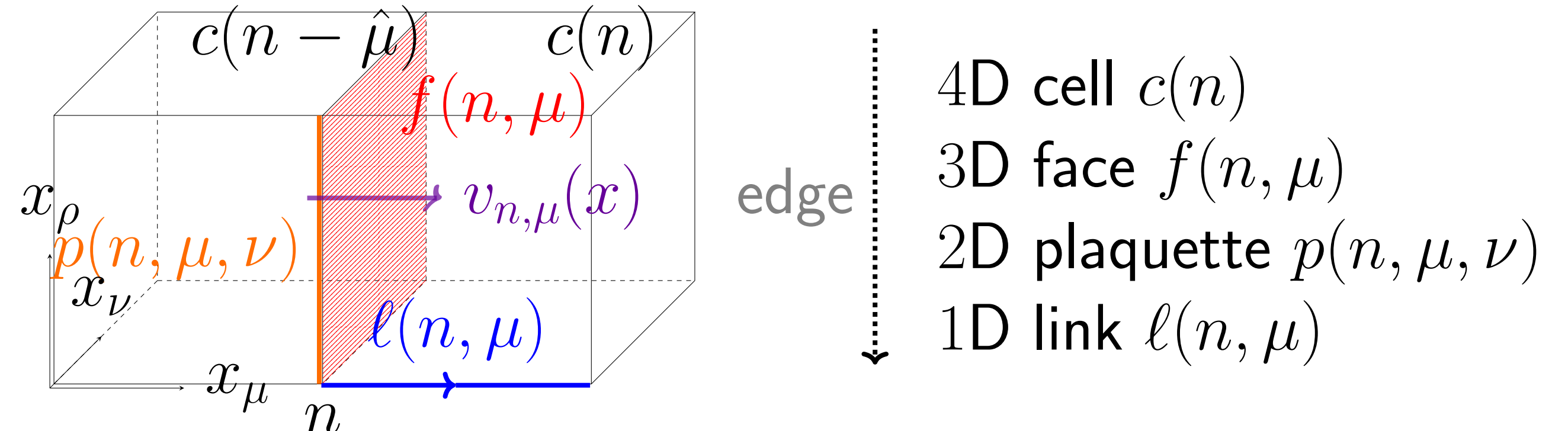
recently I released an extended package to implement B_p at

https://github.com/o-morikawa/Gaugefields.jl

- This package enables us to carry out numerical measurements with fixed B_p (**arbitrary 't Hooft flux**) or **dynamical B_p**

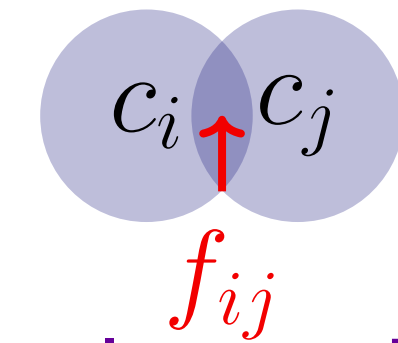
Fractional topological charge/Anomaly

- Reconstruction of fiber bundle from lattice fields [Lüscher '84]



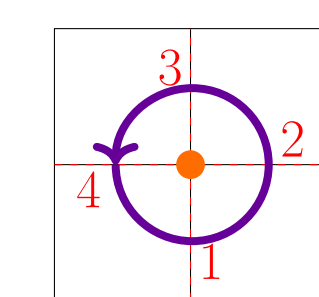
- 1 Lattice Λ divides X into 4D hypercubes; Regard $\{c(n)\}$ as patches

- 2 Define “transition function” $v_{n,\mu}(x)$ at $f(n, \mu)$ from data as u_ℓ



$$\text{Gauge transf: } u_j = v_{ij}^{-1} u_i v_{ij} \text{ at } f_{ij}$$

- cocycle condition:



$$v_1 v_2 v_3^{-1} v_4^{-1} = e^{\frac{2\pi i}{N} B_p} \text{ at } x \in p(n)$$

Admissibility condition

For uniqueness of v , we impose $\text{tr}(1 - e^{-\frac{2\pi i}{N} B_p} u_p) < \varepsilon$

- 3 Topological charge in terms of $v_\ell(x)$

$$Q = \sum_{n \in \Lambda} q(n), \quad q(n) = -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{p_{n,\mu\nu}} d^2x \text{Tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu}, \nu}^{-1} \partial_\sigma v_{n-\hat{\mu}, \nu}) \right] \right. \\ \left. + \int_{f_{n,\mu}} d^3x \text{Tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}$$

Finally, we have $Q = -\frac{\epsilon_{\mu\nu\rho\sigma} z_{\mu\nu} z_{\rho\sigma}}{8N} + \mathbb{Z}$. Then, the 't Hooft anomaly is given by $\mathcal{Z}_{\theta+2\pi}[B_p] = e^{-2\pi i Q[B_p]} \mathcal{Z}_\theta[B_p] \neq \mathcal{Z}_\theta[B_p]$

Higher group symmetry

- Gauging $G^{[0]} \times H^{[1]}$ global symmetry, if gauge transf. acts as

$$A \mapsto A + d\lambda^{(0)}, \quad B \mapsto B + d\lambda^{(1)} + \lambda^{(0)} dA$$

2-group symmetry (cf. superstring theory [Green–Schwarz '84])

- 4D $SU(N)$ gauge theory with **instanton number $p\mathbb{Z}$**

- Global symmetry: $\underbrace{\mathbb{Z}_N^{[1]} \text{ center sym}}_{\text{gauging}} \times \mathbb{Z}_p^{[3]} \text{ sym} \xrightarrow{\text{gauging}} \text{4-group}$ [Tanizaki–Ünsal '19]

- From $Q + \frac{1}{N} \sum_n \Omega_n \in p\mathbb{Z}$,

and local desc. $Q = -\frac{1}{8N} \sum_n \epsilon_{\mu\nu\rho\sigma} (B \cup B)_n + \tilde{Q}$ ($\tilde{Q} \in \mathbb{Z}$),

$$\tilde{\Omega}_n \equiv \frac{1}{N} \Omega_n - \frac{1}{8N} \sum_n \epsilon_{\mu\nu\rho\sigma} (B \cup B)_n$$

transforms under $\mathbb{Z}_N^{[1]}$ and $\mathbb{Z}_p^{[3]}$ as

$$\tilde{\Omega}_n \mapsto \tilde{\Omega}_n + d\Omega_n^{(3)} + \frac{1}{N} B \cup d\lambda + \frac{1}{N} d\lambda \cup B + \frac{1}{N} d\lambda \cup d\lambda$$

Non-invertible symmetry

- Usually, quantum anomaly $\rightarrow$ loss of symmetry

- ABJ anomaly: non-invertible symmetry for rational axial rotation [Córdova–Ohmori '22, Choi–Lam–Shao '22]

$$d \star j = \frac{1}{4\pi^2} F \wedge F \Rightarrow d(\star j - \underbrace{\frac{1}{4\pi^2} A \wedge dA}_{\text{gauge non-inv CS}}) = 0$$

Coupled to **gauge inv 3D fractional quantum Hall state** ($\sim$ frac CS)

- Problems of lattice implementation: CS theory, chiral gauge theory, anomalous gauge theory

- Instead of frac CS, for angles $\alpha = p/N$, we use level- N BF theory

$$S_{\text{BF}} = \frac{-ip\pi}{N} \sum [b(dc - z) - z \cup c]$$

where lattice fields $f = da + 2\pi z$.

- Then, for $\mathcal{Z}_{\mathcal{M}_3}[z] = \int \mathcal{D}b \mathcal{D}c e^{-S_{\text{BF}}}$, we have

$\mathcal{Z}_{\mathcal{M}_3}[z] \mathcal{Z}_{-\mathcal{M}_3}[z] = \mathcal{Z}_{\mathcal{M}_3}[0] \mathcal{C}_{\mathcal{M}_3}[z]$ where $\mathcal{C}$ is condensation operator

- We can calculate fusion rules $U_{p_1/N_1} U_{p_2/N_2} \sim U_{p_1/N_1 + p_2/N_2}$